

Model Name: P270HAN02.0

Issue Date: 2026/04/24

(*) Preliminary Specifications

() Final Specifications

Customer Signature	Date	AUO Display Plus	Date
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Record of Revision

Version	Date	Page	Description
0.1	2025/04/24	All	Preliminary datasheet release

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1. General Description

This specification applies to the 27 inch Color TFT-LCD Module P270HAN02.0. This LCD module has a TFT active matrix type liquid crystal panel 1920x1080 pixels, and diagonal size of 27 inch. This module supports 1920x1080 mode. Each pixel is divided into Red, Green and Blue sub-pixels or dots which are arranged in vertical stripes. Gray scale or the brightness of the sub-pixel color is determined with a 8-bit gray scale signal for each dot.

P270HAN02.0 has been designed to apply the 8-bit 2 channel LVDS interface method. It is intended to support displays where high brightness, wide viewing angle.

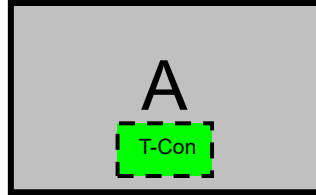
General information

Items	Specification	Unit	Note
Active Screen Size	27	inch	
Display Area	597.888(H) x 336.312(V)	mm	
Outline Dimension	623.7(H) x 362.1(V) x 12.0(D)	mm	D: front bezel to LVDS CNT
Driver Element	a-Si TFT active matrix		
Display Colors	8 bit (16.7 million)	Colors	
Number of Pixels	1920x1080	Pixel	
Pixel Pitch	0.3114 (H) x 0.3114 (W)	mm	
Pixel Arrangement	RGB vertical stripe		
Display Operation Mode	Normally Black		
Surface Treatment	Anti-Glare, SAG 25%, 3H		Haze = 25%
Rotate Function	Unachievable		Note 1
Display Orientation	Landscape/ Portrait Enable (TBD)		Note 2
Operating Time	24/7 (TBD)		
Frame Rate	60	Hz	
LED MTTF	50K	hours	

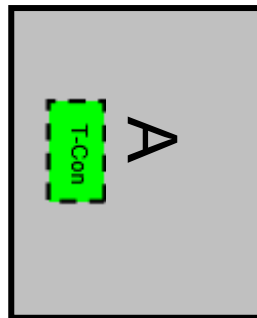
Note 1: Rotate Function refers to LCD display could be able to rotate. This function does not work in this model.

Note 2: Landscape Mode: The default placement is T-Con Side on the lower side and the image is shown upright via viewing from the front. Portrait Mode: The default placement is that T-Con side has to be placed on the left side via viewing from the front.

Landscape (Front view)



Portrait (Front view)



2. Absolute Maximum Ratings

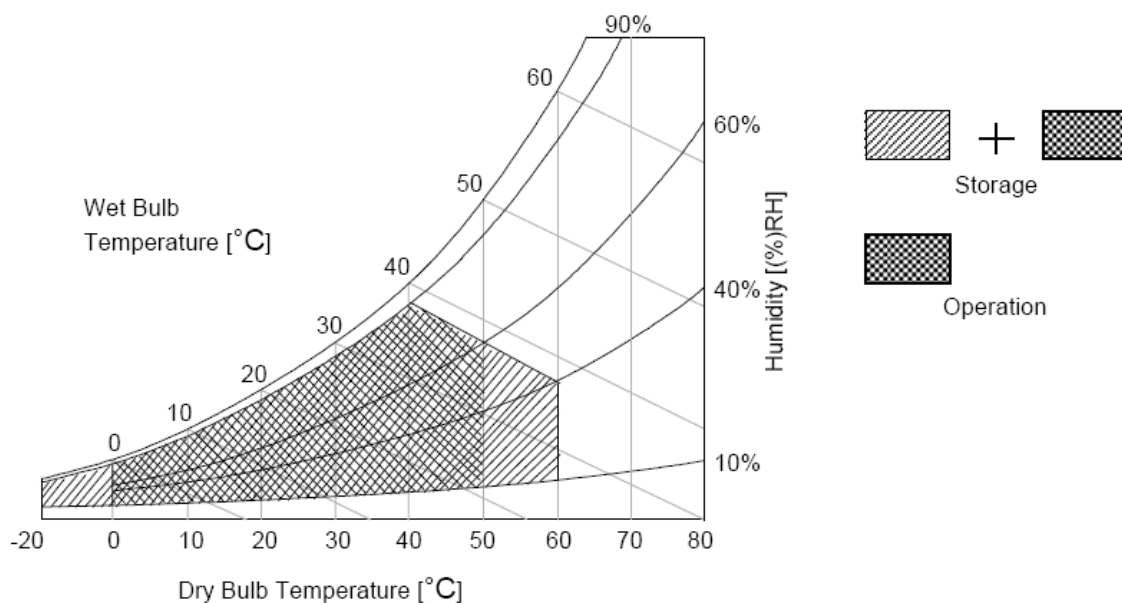
The followings are maximum values which, if exceeded, may cause faulty operation or damage to the unit.

Item	Symbol	Min	Max	Unit	Note
Logic/LCD Drive Voltage	V _{DD}	-0.3	14	[Volt]	1
Input Voltage of Signal	V _{in}	-0.3	4	[Volt]	1
Operating Temperature	TOP	-30	+70	[°C]	2
Operating Humidity	HOP	10	90	[%RH]	2
Storage Temperature	TST	-30	+80	[°C]	2
Storage Humidity	HST	10	90	[%RH]	2
Panel Surface Temperature	PST		TBD	[°C]	

Note 1: Duration:50 msec.

Note 2 : Maximum Wet-Bulb should be 39°C and No condensation.

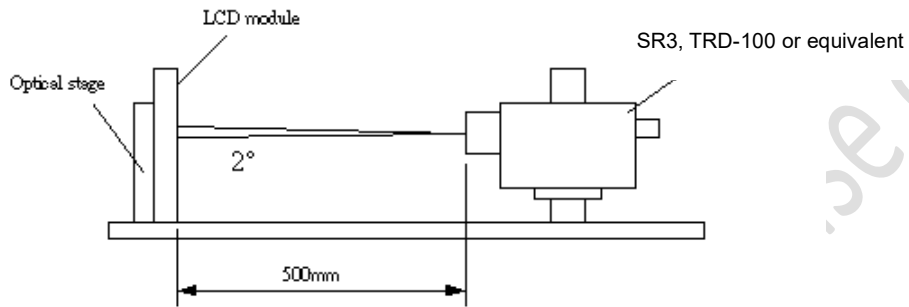
The relative humidity must not exceed 90% non-condensing at temperatures of 40°C or less. At temperatures greater than 40°C, the wet bulb temperature must not exceed 39°C.



3. Optical Specification

Optical characteristics are determined after the unit has been 'ON' and stable for approximately 45 minutes in a dark environment at 25°C. The values specified are at an approximate distance 500 mm from the LCD surface at a viewing angle of φ and θ equal to 0°.

Fig.1 presents additional information concerning the measurement equipment and method.



Parameter		Symbol	Values			Unit	Note
			Min.	Typ.	Max		
Contrast Ratio		CR	600	1000	--		1
Surface Luminance (White)		L _{WH}	1040	1300	--	cd/m ²	2
Luminance Variation		$\delta_{\text{WHITE(9P)}}$	--	--	1.33		3
Response Time (G to G)		T _Y	--	14		ms	4
Color Gamut		NTSC		72		%	
Color Coordinates							
	Red	R _X	Typ.-0.03	0.651	Typ.+0.03		
		R _Y		0.329			
	Green	G _X		0.304			
		G _Y		0.616			
	Blue	B _X		0.152			
		B _Y		0.065			
	White	W _X		0.329			
		W _Y		0.353			
Viewing Angle							5
	x axis, right($\varphi=0^\circ$)	θ_r	85	89	--	degree	
	x axis, left($\varphi=180^\circ$)	θ_l	85	89	--	degree	
	y axis, up($\varphi=90^\circ$)	θ_u	85	89	--	degree	
	y axis, down ($\varphi=270^\circ$)	θ_d	85	89	--	degree	

Note 1: Contrast Ratio (CR) is defined mathematically as:

$$\text{Contrast Ratio} = \frac{\text{Surface Luminance of } L_{\text{on5}}}{\text{Surface Luminance of } L_{\text{off5}}}$$

Note 2: Surface luminance is luminance value at point 5 across the LCD surface 50cm from the surface with all pixels displaying white. From more information see FIG 2. LED current I_F = typical value (without driver board), LED input $V_{\text{DDB}} = 24\text{V}$, I_{DDB} = Typical value (with driver board), $L_{\text{WH}} = L_{\text{on5}}$ where L_{on5} is the luminance with all pixels displaying white at center 5 location.

Note 3: The variation in surface luminance, δWHITE is defined (center of Screen) as:

$$\delta\text{WHITE(9P)} = \frac{\text{Maximum}(L_{\text{on1}}, L_{\text{on2}}, \dots, L_{\text{on9}})}{\text{Minimum}(L_{\text{on1}}, L_{\text{on2}}, \dots, L_{\text{on9}})}$$

Note 4: Response time T_γ is the average time required for display transition by switching the input signal for five luminance ratio (0%,25%,50%,75%,100% brightness matrix) and is based on Frame rate = 60Hz to optimize.

Measured Response Time		Target				
		0%	25%	50%	75%	100%
Start	0%		0% to 25%	0% to 50%	0% to 75%	0% to 100%
	25%	25% to 0%		25% to 50%	25% to 75%	25% to 100%
	50%	50% to 0%	50% to 25%		50% to 75%	50% to 100%
	75%	75% to 0%	75% to 25%	75% to 50%		75% to 100%
	100%	100% to 0%	100% to 25%	100% to 50%	100% to 75%	

T_γ is determined by 10% to 90% brightness difference of rising or falling period. (As illustrated)

The response time is defined as the following figure and shall be measured by switching the input signal for “any level of gray(bright)” and “any level of gray(dark)”.

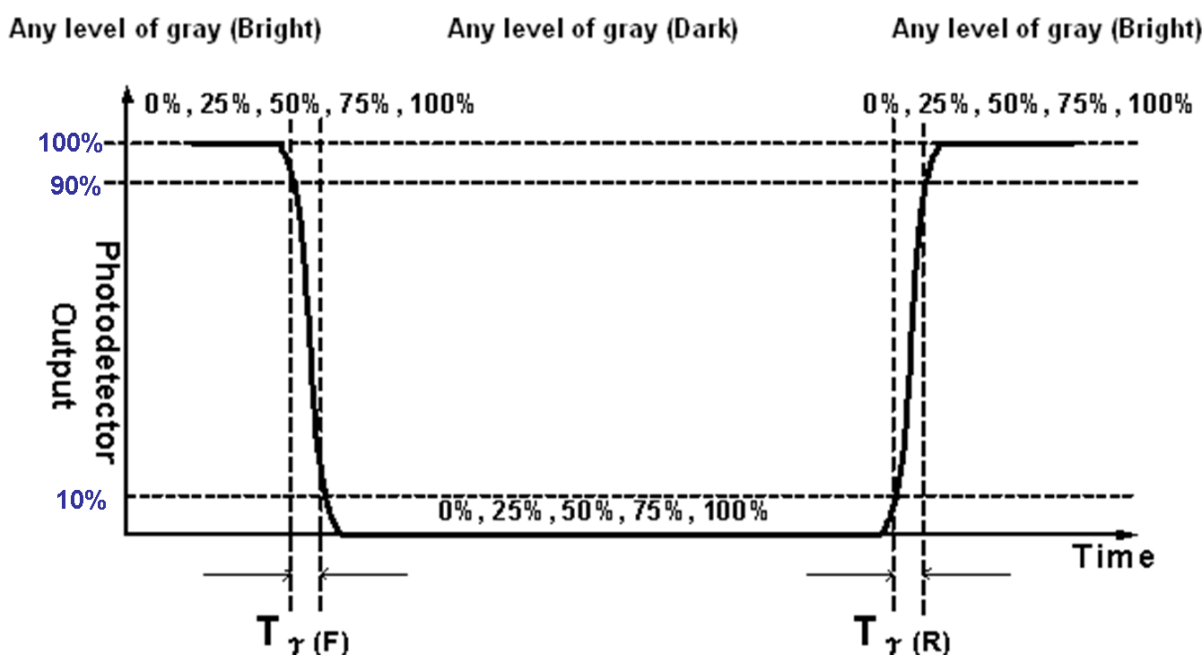
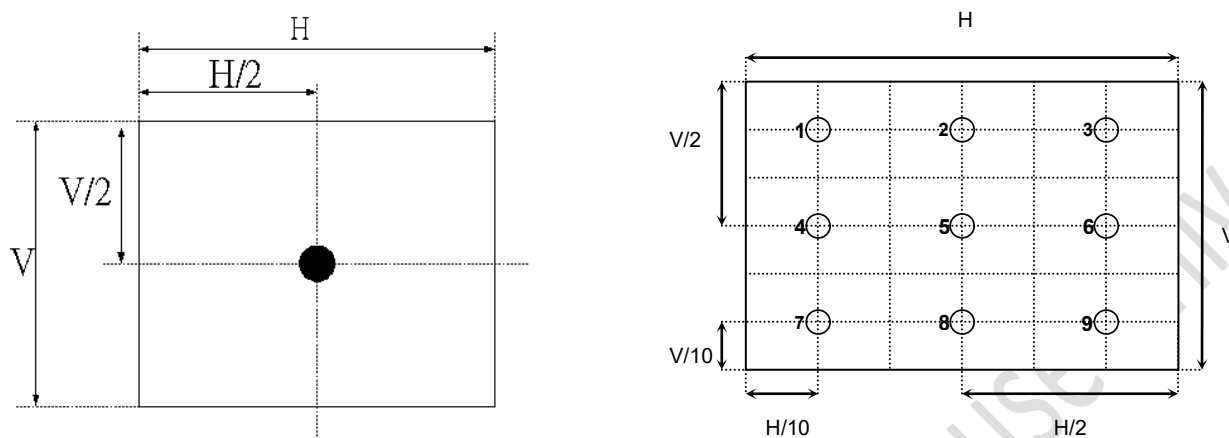
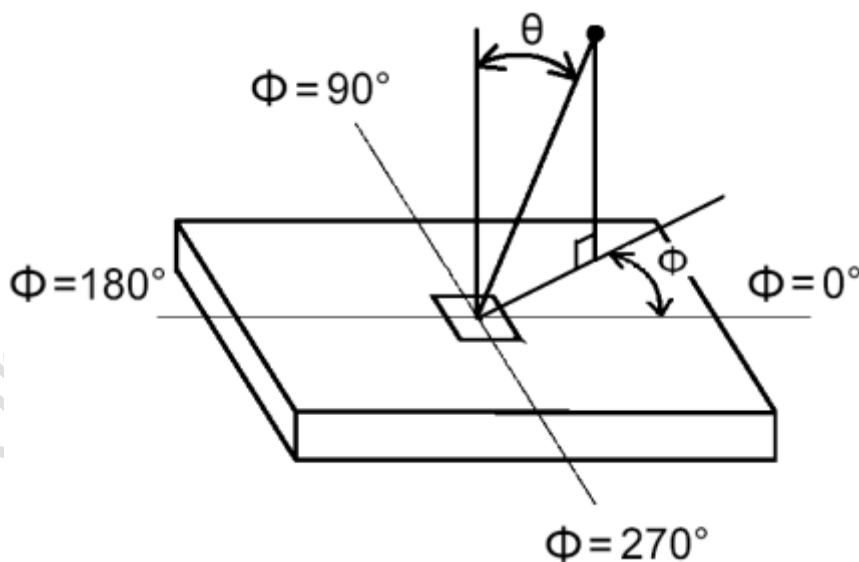


FIG. 2 Luminance



- 5 Viewing angle is the angle at which the contrast ratio is greater than 10. The angles are determined for the horizontal or x axis and the vertical or y axis with respect to the z axis which is normal to the LCD surface. For more information see FIG3.

FIG.3 Viewing Angle



4. Interface Specification

4.1 Input power

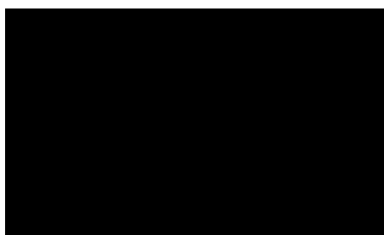
The P270HVN04.1 module requires power inputs which are employed to power the LCD electronics and to drive the TFT array and liquid crystal.

Item	Symbol	Min.	Typ.	Max	Unit	Note
Power Supply Input Voltage	V_{DD}	4.5	5.0	5.5	V	1
Power Supply Input Current (White Pattern)	I_{DD}		0.37	0.85	A	2
Power Consumption (White Pattern)	P_{DD}		1.85	4.25	Watt	2
Inrush Current	I_{RUSH}	--	--	3.0	A	3

Note 1: The ripple voltage should be fewer than 5% of V_{DD} .

Note 2: Test Condition: (1) $V_{DD} = 5.0V$, (2) $F_v = 60Hz$, (3) $F_{clk} = 74.25MHz$, (4) Temperature = 25 °C (5) Power dissipation check pattern. (Only for power design)

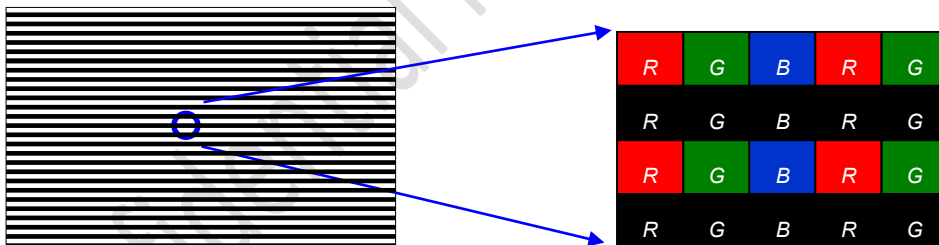
a. Black pattern



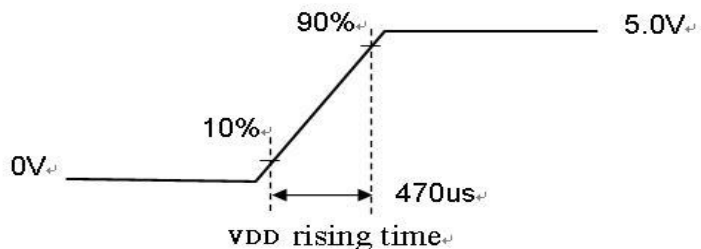
b. White pattern



c. H-Strip pattern



Note 3: The duration of V_{DD} rising time: 470us



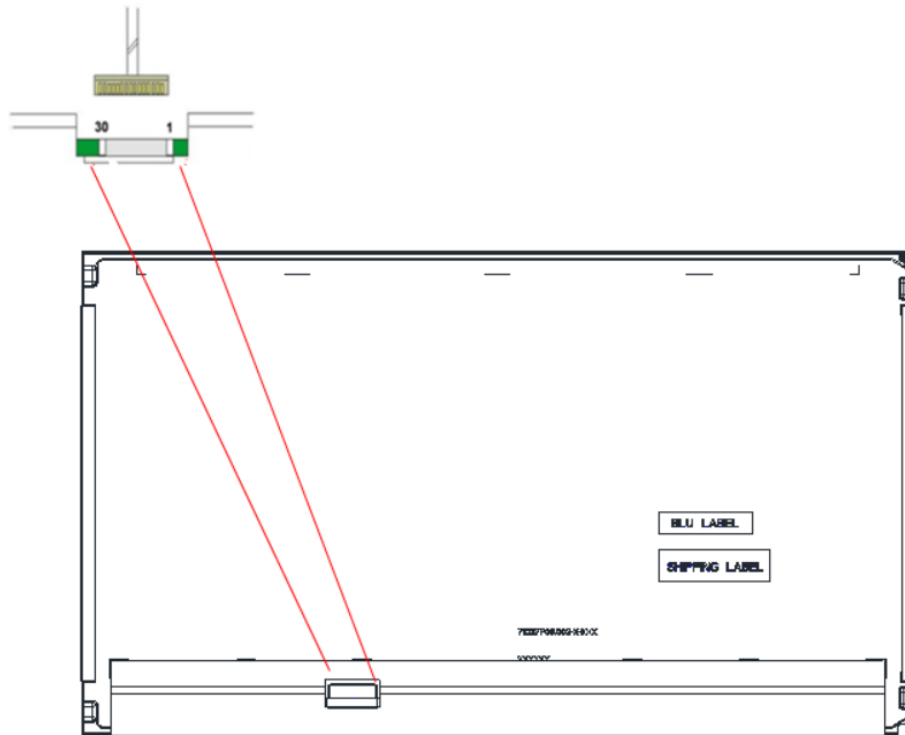
4.2 Input Connection

TFT-LCD connector: (STM) MSBKT2407P30HB、(P-TWO)187034-3009

Mating Connector: (JAE) FI-X30HL or Compatible

PIN	Symbol	Description	Note
1	RxO0-	Negative LVDS differential data input (Odd data)	1
2	RxO0+	Positive LVDS differential data input (Odd data)	
3	RxO1-	Negative LVDS differential data input (Odd data)	
4	RxO1+	Positive LVDS differential data input (Odd data)	
5	RxO2-	Negative LVDS differential data input (Odd data)	
6	RxO2+	Positive LVDS differential data input (Odd data)	
7	GND	Ground	
8	RxOCLK-	Negative LVDS differential clock input (Odd clock)	
9	RxOCLK+	Positive LVDS differential clock input (Odd clock)	
10	RxO3-	Negative LVDS differential data input (Odd data)	
11	RxO3+	Positive LVDS differential data input (Odd data)	
12	RxE0-	Negative LVDS differential data input (Even data)	
13	RxE0+	Positive LVDS differential data input (Even data)	
14	GND	Ground	
15	RxE1-	Negative LVDS differential data input (Even data)	
16	RxE1+	Positive LVDS differential data input (Even data)	
17	GND	Ground	
18	RxE2-	Negative LVDS differential data input (Even data)	
19	RxE2+	Positive LVDS differential data input (Even data)	
20	RxECLK-	Negative LVDS differential clock input (Even clock)	
21	RxECLK+	Positive LVDS differential clock input (Even clock)	
22	RxE3-	Negative LVDS differential data input (Even data)	
23	RxE3+	Positive LVDS differential data input (Even data)	
24	GND	Ground	
25	NC	No connection (for AUO test only. Do not connect)	2
26	NC	No connection (for AUO test only. Do not connect)	2
27	NC	No connection (for AUO test only. Do not connect)	2
28	VDD	Power Supply Input Voltage	
29	VDD	Power Supply Input Voltage	
30	VDD	Power Supply Input Voltage	

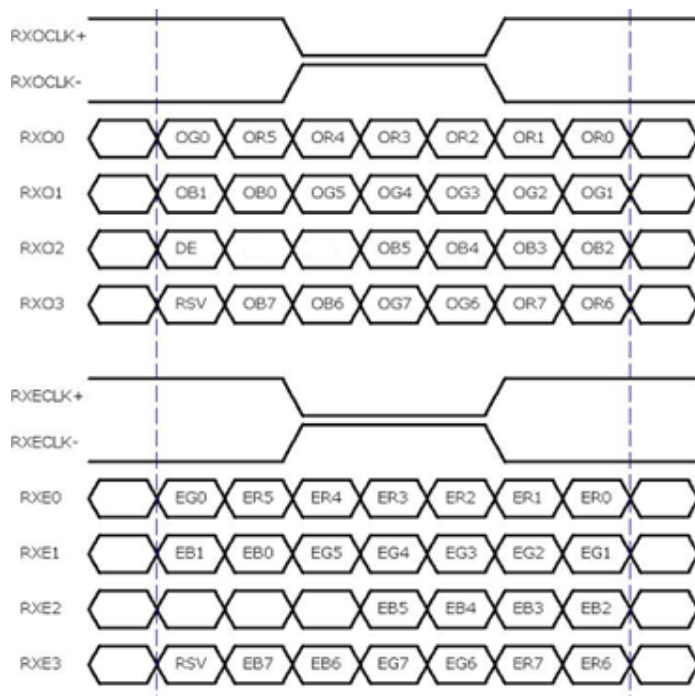
Note 1: Pin assignment as the following figure



Note 2: Please leave this pin unoccupied. It cannot be connected with any signal (Low/GND/High).

4.3 Input Data Format

4.3.1 LVDS Data Format



8 Bit Color Bit Order			
MSB	R7	G7	B7
	R6	G6	B6
	R5	G5	B5
	R4	G4	B4
	R3	G3	B3
	R2	G2	B2
	R1	G1	B1
LSB	R0	G0	B0

Note: O = "Odd Pixel Data", E = "Even Pixel Data". Refer to 3.4.1 LCD pixel format, the 1st data is 1 (Odd Pixel Data), the 2nd data is 2 (Even Pixel Data) and the last data is 1920 (Even Pixel Data).

4.3.2 Color versus Input data

The brightness of each primary color (red, green and blue) is based on the 8 bit gray scale data input for the color; the higher the binary input, the brighter the color. The table below provides a reference for color versus data input.

Color		Input Color Data																							
		RED								GREEN								BLUE							
		(MSB: R7, LSB: R0)								(MSB: G7, LSB: G0)								(MSB: B7, LSB: B0)							
		R7	R6	R5	R4	R3	R2	R1	R0	G7	G6	G5	G4	G3	G2	G1	G0	B7	B6	B5	B4	B3	B2	B1	B0
Basic Color	Black	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Red(255)	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Green(255)	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0
	Blue(255)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1
	Cyan	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	Magenta	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1
	Yellow	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0
	White	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
R	RED(000)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	RED(001)	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	RED(254)	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	RED(255)	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
G	GREEN(000)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	GREEN(001)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0

	GREEN(254)	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0
	GREEN(255)	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0
B	BLUE(000)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	BLUE(001)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

	BLUE(254)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	0
	BLUE(255)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1

5. Signal Timing Specification

This is the signal timing required at the input of the user connector. All of the interface signal timing should be satisfied with the following specifications for its proper operation.

Timing Table (DE only Mode)

Signal	Item	Symbol	Min.	Typ.	Max	Unit
Vertical Section	Period	Tv	1094	1130	1914	Th
	Active	Tdisp (v)	1080			2160
	Blanking	Tblk (v)	14	50	834	Th
	Frequency	Fv	47	60	76	Hz
Horizontal Section	Period	Th	1000	1050	1678	Tclk
	Active	Tdisp (h)	960			
	Blanking	Tblk (h)	40	90	718	Tclk
	Frequency	Fh	51.5	67.8	90.0	KHz
LVDS Clock	Frequency	Fclk=1/Tclk	51.5	71.2	90	MHz

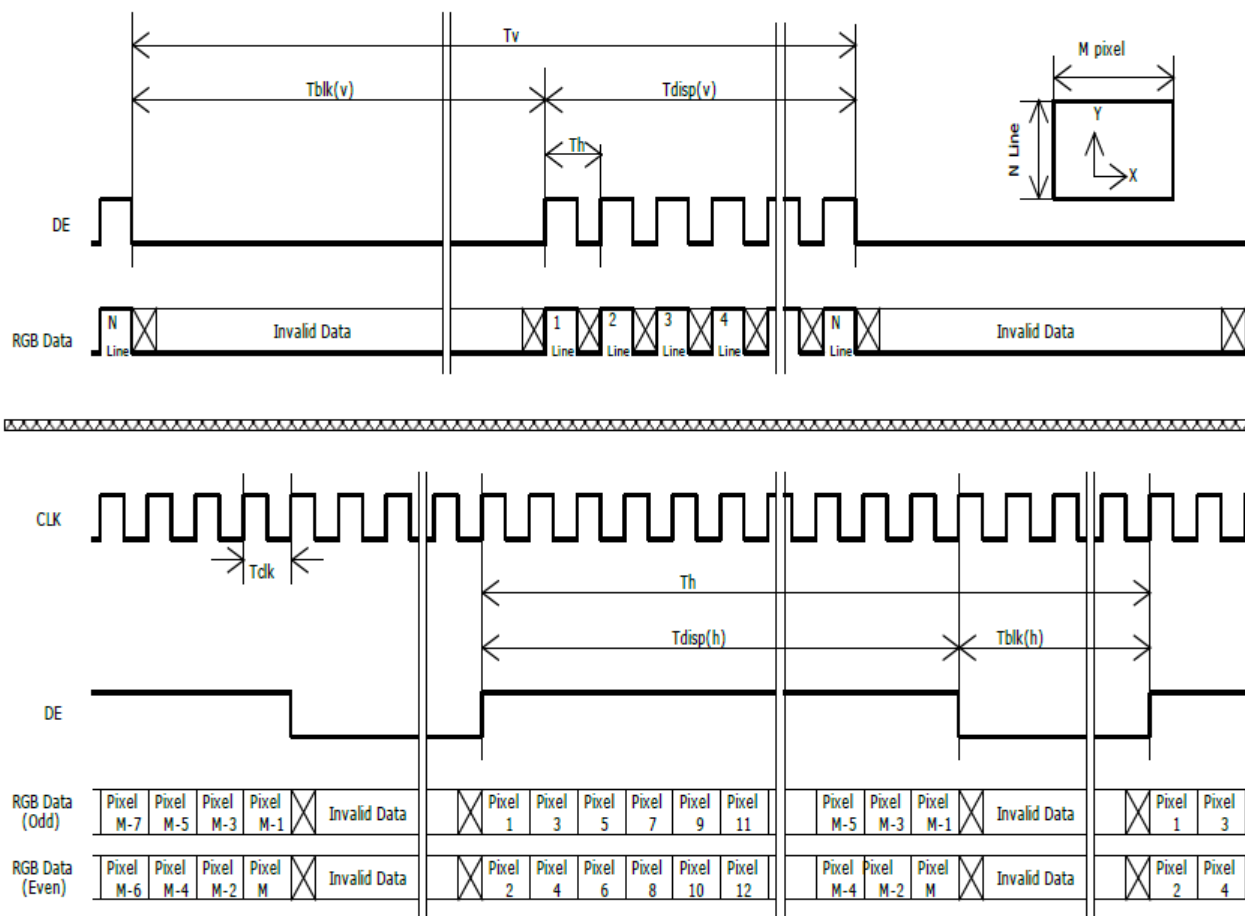
Note 1: Display position is specific by the rise of DE signal only. Horizontal display position is specified by the rising edge of 1st DCLK after the rise of 1st DE, is displayed on the left edge of the screen.

Note 2: Vertical display position is specified by the rise of DE after a “Low” level period equivalent to eight times of horizontal period. The 1st data corresponding to one horizontal line after the rise of 1st DE is displayed at the top line of screen.

Note 3: If a period of DE “High” is less than 1920 DCLK or less than 1080 lines, the rest of the screen displays black.

Note 4: The display position does not fit to the screen if a period of DE “High” and the effective data period do not synchronize with each other.

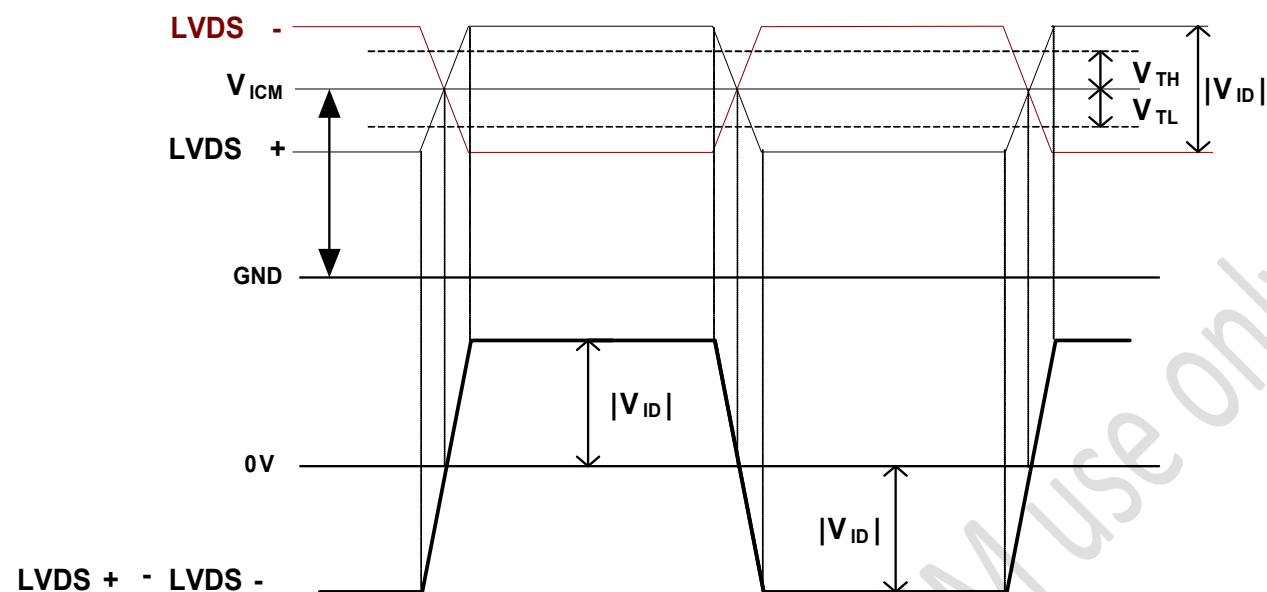
5.1 Signal Timing Waveform



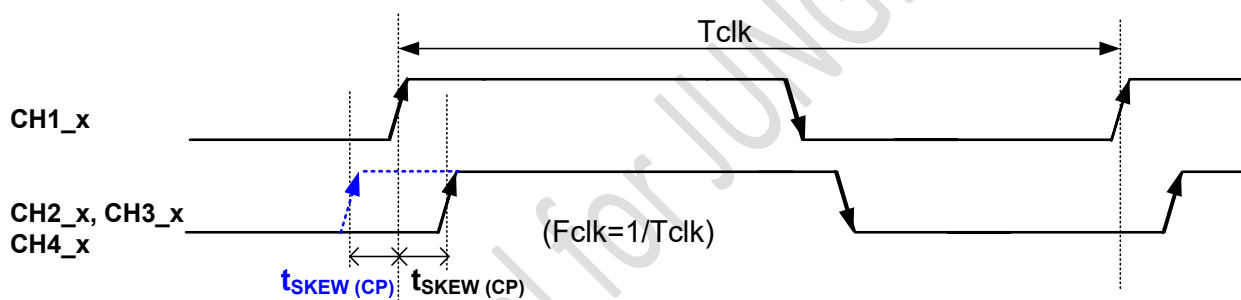
5.2 Input Timing Specification

Parameter		Symbol	Value			Unit	Note
			Min.	Typ.	Max		
LVDS Interface	Input Differential Voltage	$ V_{ID} $	100	-	600	mV _{DC}	1
	Differential Input High Threshold Voltage	V_{TH}	-	-	100	mV _{DC}	1
	Differential Input Low Threshold Voltage	V_{TL}	-100	-	-	mV _{DC}	1
	Input Common Mode Voltage	V_{ICM}	1.0	1.2	1.5	V _{DC}	1
	Input Channel Pair Skew Margin	$t_{SKEW (CP)}$	-500	--	+500	ps	2
	Input Channel Pair Skew Margin (only for M'Star MST7428BB)	$t_{SKEW (CP)}$	-400	--	+400	ps	2
	Receiver Clock : Spread Spectrum Modulation range	F_{clk_ss}	$F_{clk} - 3\%$	--	$F_{clk} + 3\%$	MHz	3
	Receiver Clock : Spread Spectrum Modulation frequency	F_{ss}	30	--	200	KHz	3
	Receiver Data Input Margin $F_{clk} = 85 \text{ MHz}$ $F_{clk} = 65 \text{ MHz}$	t_{RMG}	-0.4 -0.5	-- --	0.4 0.5	ns	8

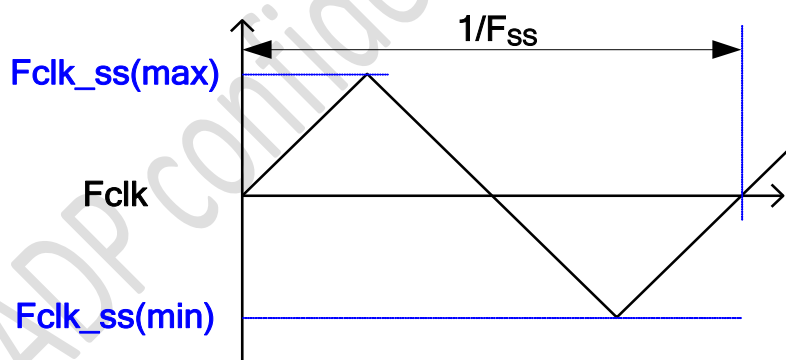
Note 1: $V_{ICM} = 1.2V$



Note 2: Input Channel Pair Skew Margin

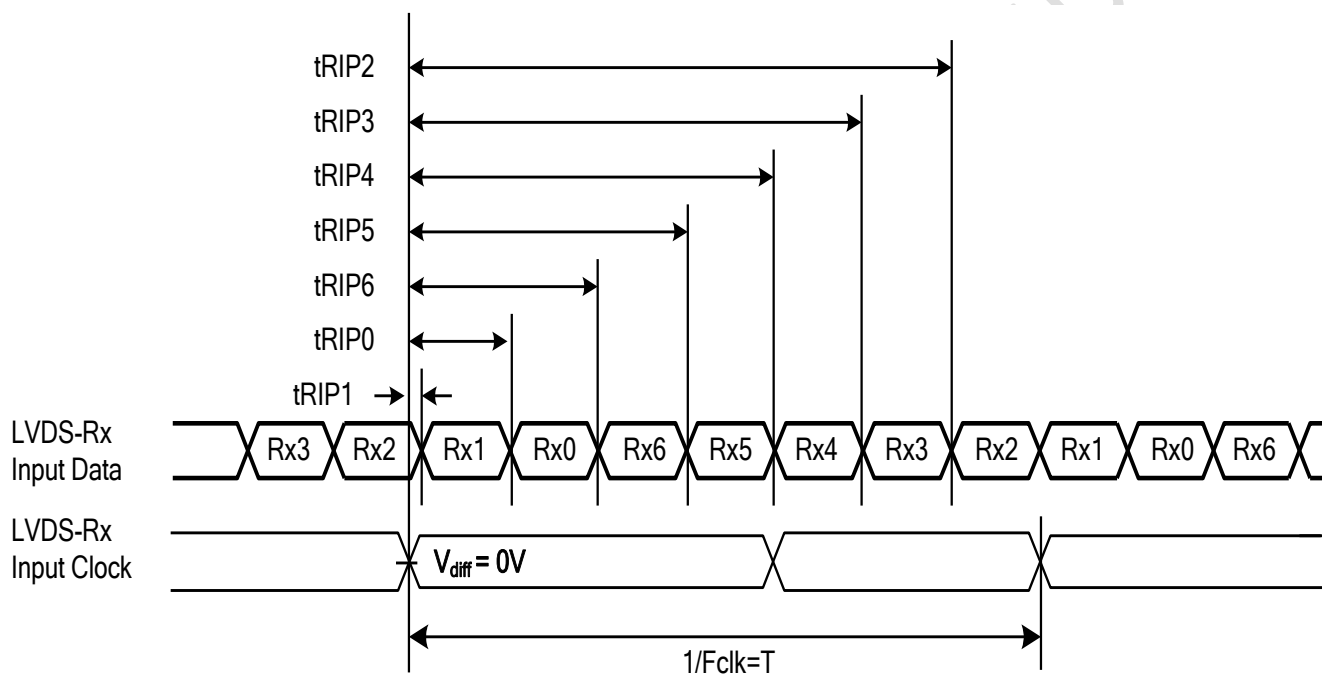


Note 3: LVDS Receiver Clock SSCG (Spread spectrum clock generator) is defined as below figures.

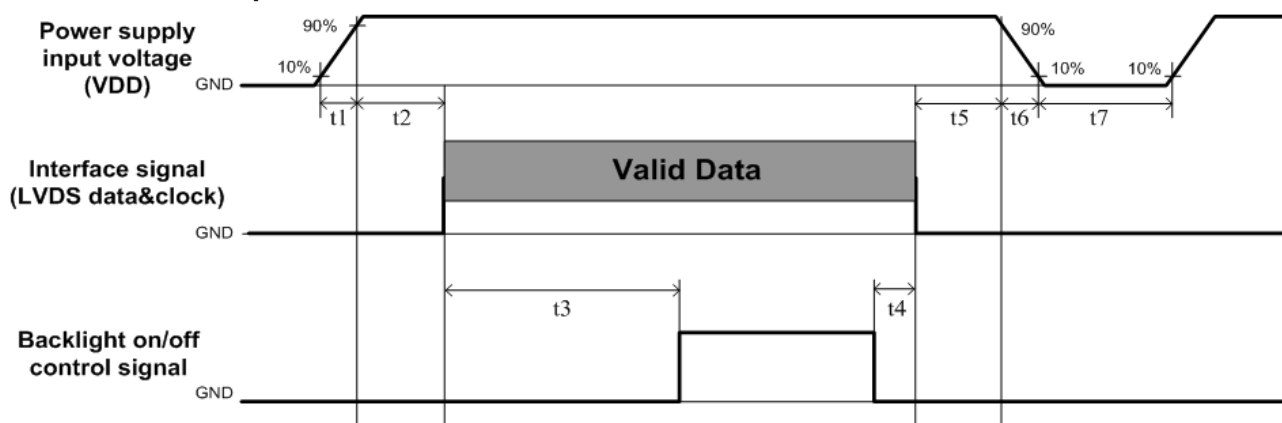


Note 4: Receiver Data Input Margin

Parameter	Symbol	Rating			Unit	Note
		Min	Type	Max		
Input Clock Frequency	Fclk	Fclk (min)	-	Fclk (max)	MHz	$T=1/Fclk$
Input Data Position0	tRIP1	- tRMG	0	tRMG	ns	
Input Data Position1	tRIP0	$T/7- tRMG $	$T/7$	$T/7+ tRMG $	ns	
Input Data Position2	tRIP6	$2T/7- tRMG $	$2T/7$	$2T/7+ tRMG $	ns	
Input Data Position3	tRIP5	$3T/7- tRMG $	$3T/7$	$3T/7+ tRMG $	ns	
Input Data Position4	tRIP4	$4T/7- tRMG $	$4T/7$	$4T/7+ tRMG $	ns	
Input Data Position5	tRIP3	$5T/7- tRMG $	$5T/7$	$5T/7+ tRMG $	ns	
Input Data Position6	tRIP2	$6T/7- tRMG $	$6T/7$	$6T/7+ tRMG $	ns	



5.3 Power Sequence for LCD



Parameter	Values			Unit
	Min.	Type.	Max.	
t1	0.5	-	10	ms
t2	0	-	50	ms
t3	500	-	-	ms
t4	100	-	-	ms
t5	0	-	50	ms
t6	0	-	200	ms
t7	1000	-	-	ms

Note 1: t4=0 : concern for residual pattern before BLU turn off.

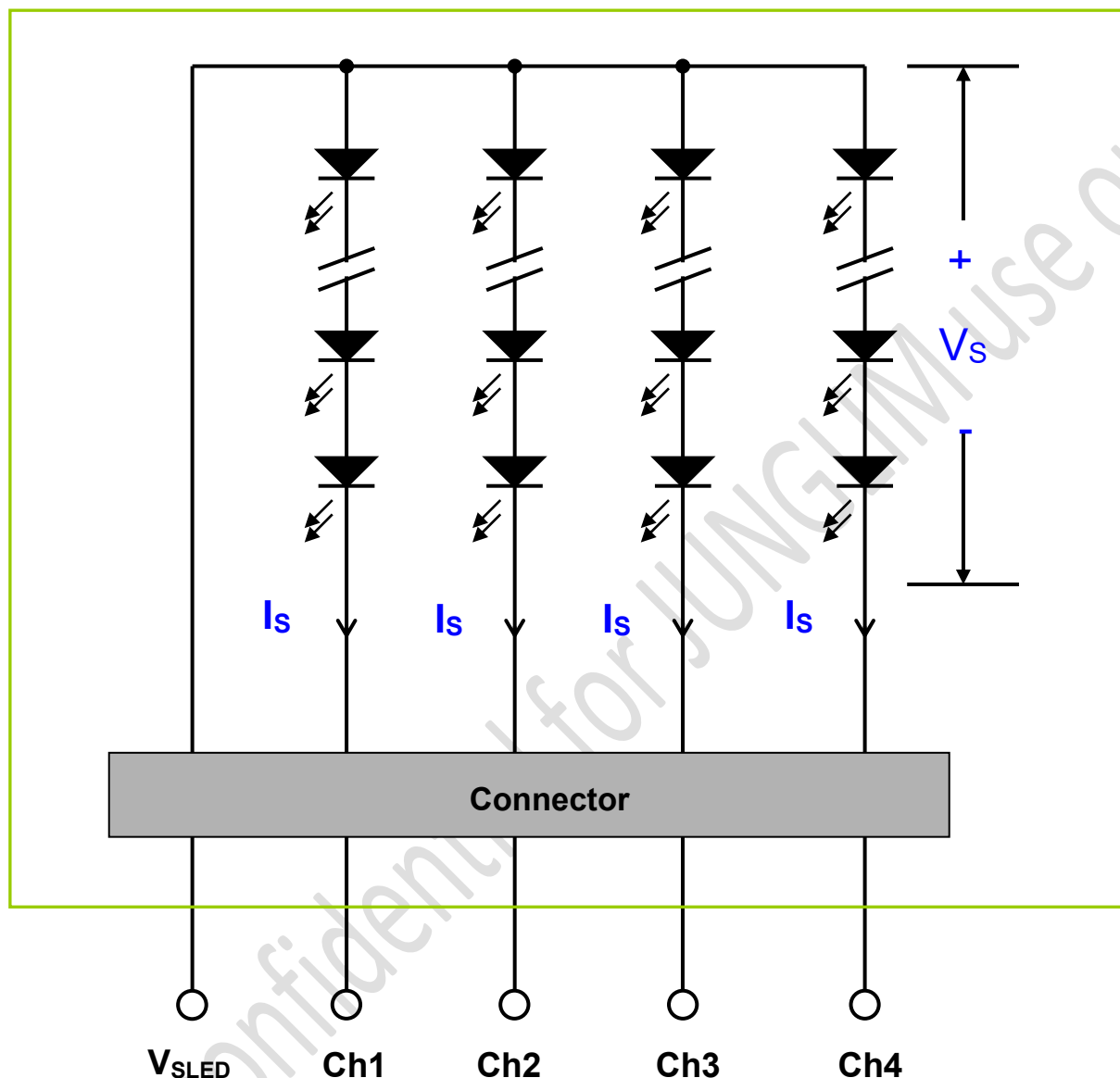
Note 2: t6 : voltage of VDD must decay smoothly after power-off. (customer system decide this value)

Note 3: t7 : When the power supply input voltage(VDD) is off, be sure to pull down the valid and invalid data to 0V.

6. Backlight Specification

6.1 Block Diagram

The following shows the block diagram of the 27 inch Backlight Unit. And it includes 116pcs LED in one LED light bar. This module has two LED light bars. (4 strings and 29 pcs LED of one string).



6.2 Interface Connection

6.2.1 Connector Type

Backlight Connector	Manufacturer	CVILUX
	Part Number	CI1406M1HRN-NH1
Mating Connector	Manufacturer	CVILUX or Compatible
	Part Number	CI1406SL000-NH (Lock)

6.2.2 Connector Pin Assignment

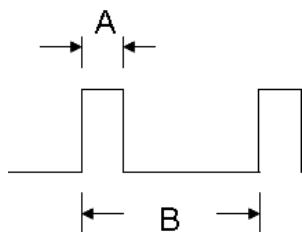
Pin#	Symbol	Description
1	Ch1	LED Current Feedback Terminal (Channel 1)
2	Ch2	LED Current Feedback Terminal (Channel 2)
3	V _{SLED}	LED Power Supply Voltage Input Terminal
4	V _{SLED}	LED Power Supply Voltage Input Terminal
5	Ch3	LED Current Feedback Terminal (Channel 3)
6	Ch4	LED Current Feedback Terminal (Channel 4)

6.3 Electrical Characteristics

6.3.1 Absolute Maximum Rating

Permanent damage may occur if exceeding the following maximum rating. (Ta=25°C)

Symbol	Description	Min	Max	Unit	Remark
Is	LED String Current	0	TBD	[mA]	100% duty ratio
			TBD	[mA]	<i>Duty ratio ≤ 10%</i> <i>Pulse time=10 ms</i>



Duty ratio= (A / B) X 100% ; (A: Pulse time, B: Period)

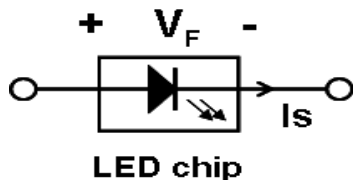
6.3.2 Recommended Operating Condition(Ta=25°C)

Symbol	Description	Min.	Typ.	Max.	Unit	Remark
Is	LED String Current	-	61		[mA]	100% duty ratio of LED chip Note 7
Vs	LED String Voltage		81.2	87.87	[Volt]	Is=61mA @ 100% duty ratio; Note 1, Note 5
ΔVs	Maximum Vs Voltage Deviation of light bar	-	-	4.0	[Volt]	Is=61mA @ 100% duty ratio; Note 2
P _{BLU}	LED Light Bar Power Consumption	-	39.7	42.9	[Watt]	Note 3
LT _{LED}	LED MTTF	50,000		-	[Hour]	Note 4
OVP	Over Voltage Protection in system board	110% Vsmax	-	-	[Volt]	Note 5

Note 1: Vs (Typ.) = V_F (Typ.) X LED No. (one string);

a. V_F: LED chip forward voltage, V_F (Min.)=2.48 V_F(Typ.)=2.81, V_F(Max.)=2.88V

b. The same equation to calculate Vs(Min.) & Vs (Max.) for respective V_F (Min.) & V_F(Max.);



Note 2: ΔVs (Max.) = ΔV_F X LED No. (one string);

a. ΔV_F: LED chip forward voltage deviation; (0.2 V , each Bin of LED V_F)

Note 3: P_{BLU} (Typ.) = Vs (Typ.) X Is (Typ.) X 4 ; (4 is total String No. of LED Light bar), P_{BLU} (Max.) = Vs (Max.) X Is (Typ.) X 4 ;

Note 4: LED MTTF is defined as the time which luminance of LED is 50% compared to its original value.[Operating condition: Continuous operating at Ta = 25±2°C, for single LED only] MTTF is a reference index, it is not representative of warranty.

Note 5: Recommendation for LED driver power design: Due to there are electrical property deviation in LED & monitor set system component after long time operation. ADP strongly recommend the design value of LED driver board OVP (over voltage protection) should be 10% higher than max. value of LED string voltage (Vs) at least.

Note 6: ADP strongly recommend “Analog Dimming” method for backlight brightness control for Wavy Noise Free. Otherwise, recommend that Dimming Control Signal (PWM Signal) should be synchronized with Frame Frequency.

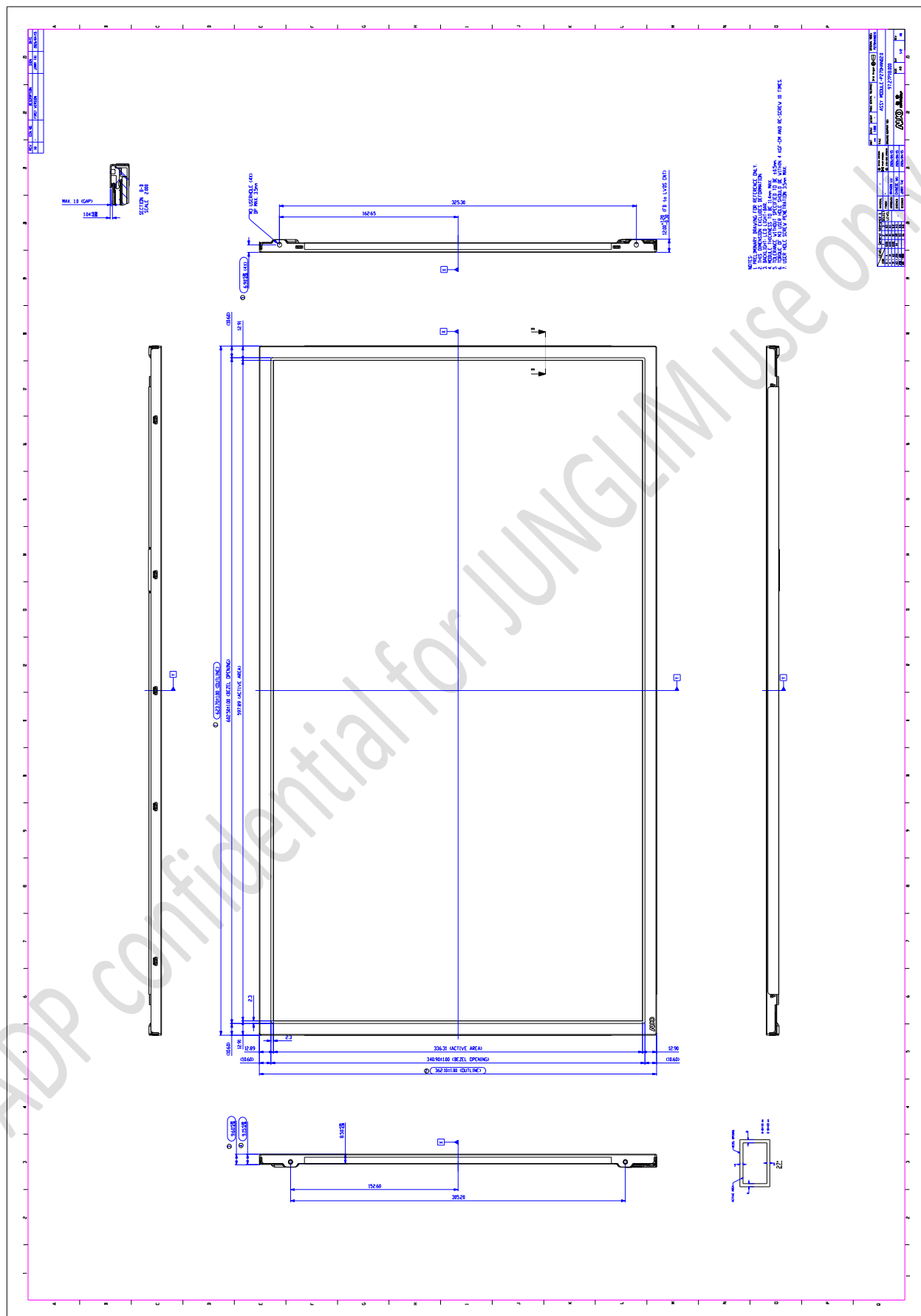
Note 7: Ensure that the LED light bar is not subjected either forward or reverse voltage while monitor set is on standby mode or not in use.

7. Mechanical Characteristics

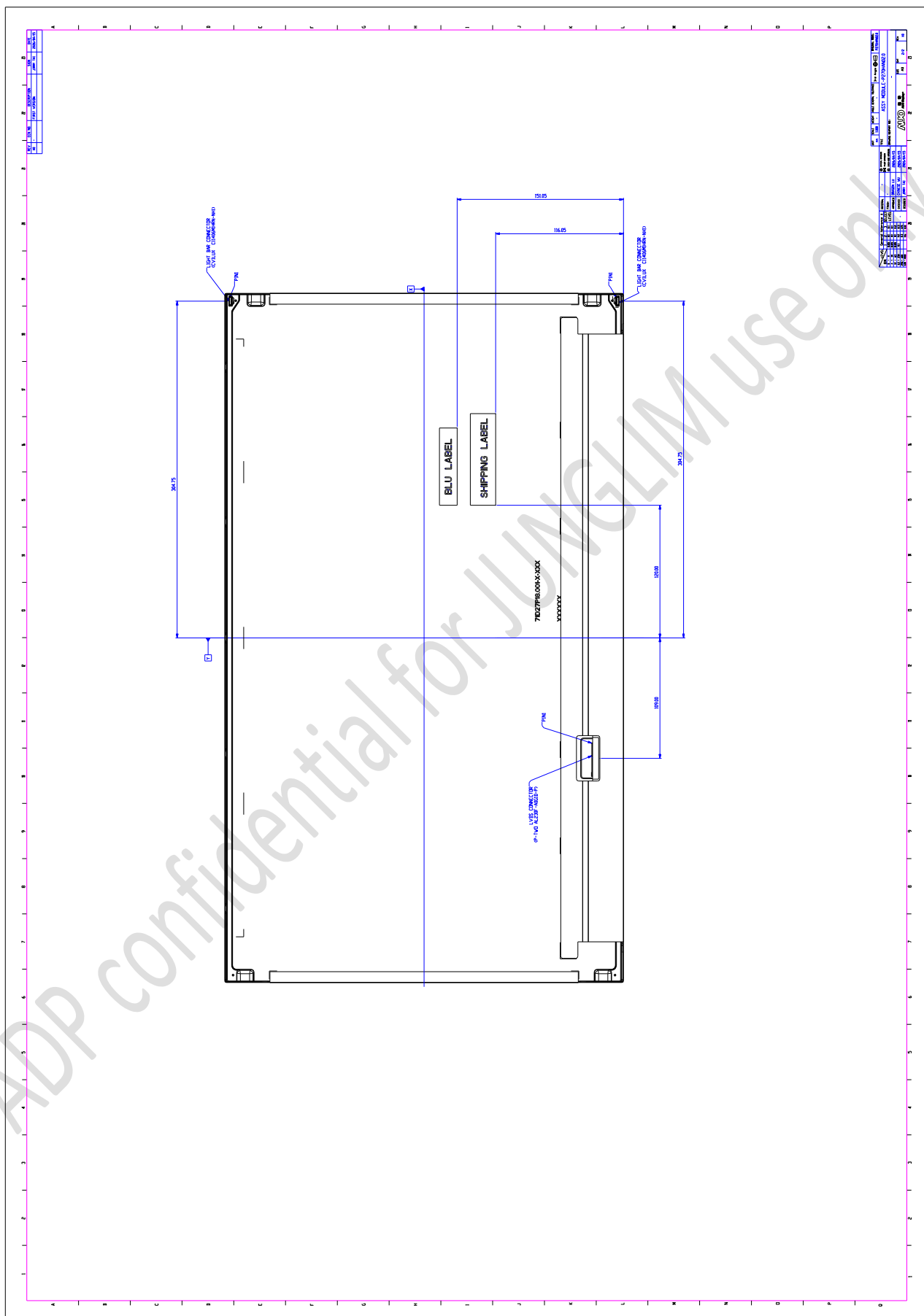
The contents provide general mechanical characteristics for the model P270HAN02.0. In addition the figures in the next page are detailed mechanical drawing of the LCD.

Item		Dimension	Unit	Note
Outline Dimension	Horizontal	623.7	mm	
	Vertical	362.1	mm	
	Depth (Dmin)	8.5	mm	Front bezel to Back Bezel
	Depth (Dmax)	12.0	mm	Front Bezel to SPCB CN
	Bezel opening	602.5(H) x 340.9(V)	mm	
	Bezel Width	10.6/10.6/10.6/10.6	mm	U/D/L/R
	Display Area	597.89(H) x 336.31(V)	mm	
Weight	3.23		Kg	

Front View



Back View



8. Reliability Test Items

Test Item	Q'ty	Condition
High temperature storage test	3	80°C, 500hrs
Low temperature storage test	3	-30°C, 500hrs
High temperature operation test	3	70°C, 500hrs
Low temperature operation test	3	-30°C, 500hrs
Vibration test (With carton)	1(PKG)	Random wave (1.5Grms 10~200Hz) Duration : X,Y,Z 20min per axes
Drop test (With carton)	1(PKG)	Height: 45.7 cm Direction: Only bottom flat twice (ASTMD4169-I)

9. International Standard

9.1 Safety

- (1) UL 62368-1; Audio/video, information and communication technology equipment – Part 1: Safety requirements.
- (2) IEC 62368-1; Audio/video, information and communication technology equipment – Part 1: Safety requirements.
- (3) EN 62368-1; Audio/video, information and communication technology equipment – Part 1: Safety requirements.

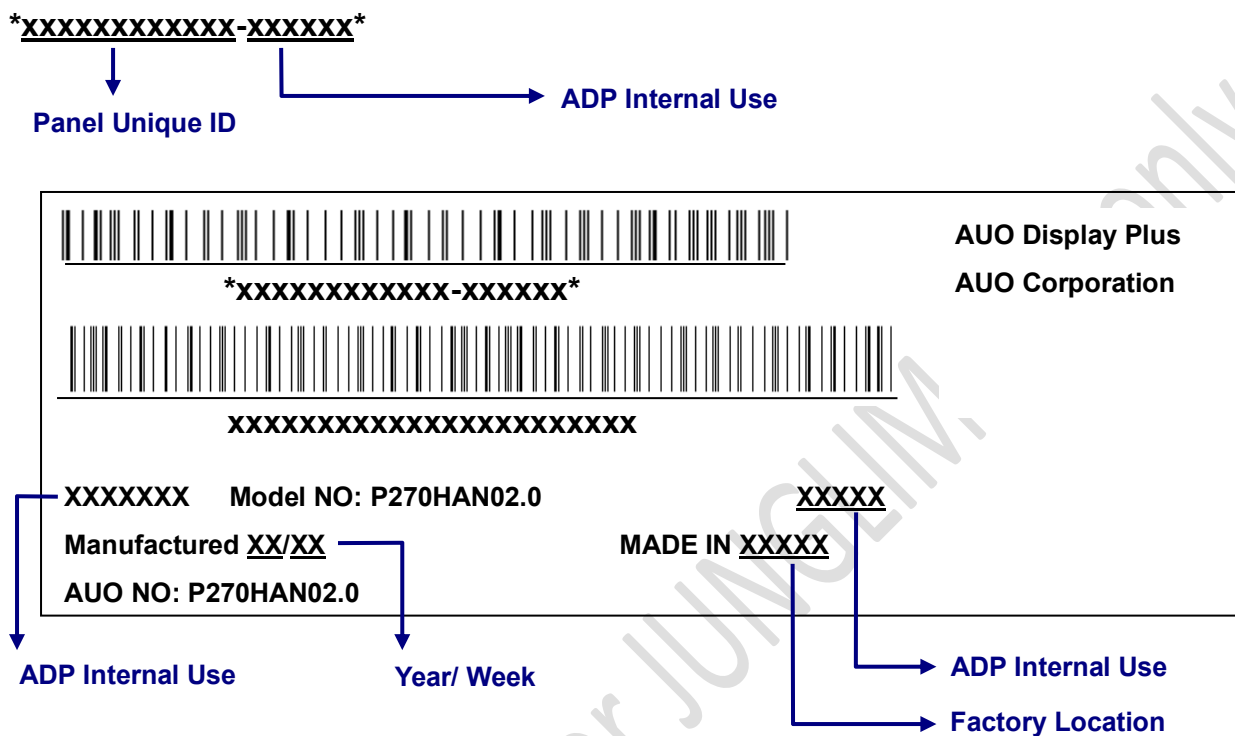
9.2 EMC

- (1) ANSI C63.4 “Methods of Measurement of Radio-Noise Emissions from Low-Voltage Electrical and Electrical Equipment in the Range of 9kHz to 40GHz. “American National standards Institute(ANSI), 1992
- (2) C.I.S.P.R “Limits and Methods of Measurement of Radio Interface Characteristics of Information Technology Equipment.” International Special committee on Radio Interference.
- (3) EN 55022 “Limits and Methods of Measurement of Radio Interface Characteristics of Information Technology Equipment.” European Committee for Electrotechnical Standardization. (CENELEC), 1998

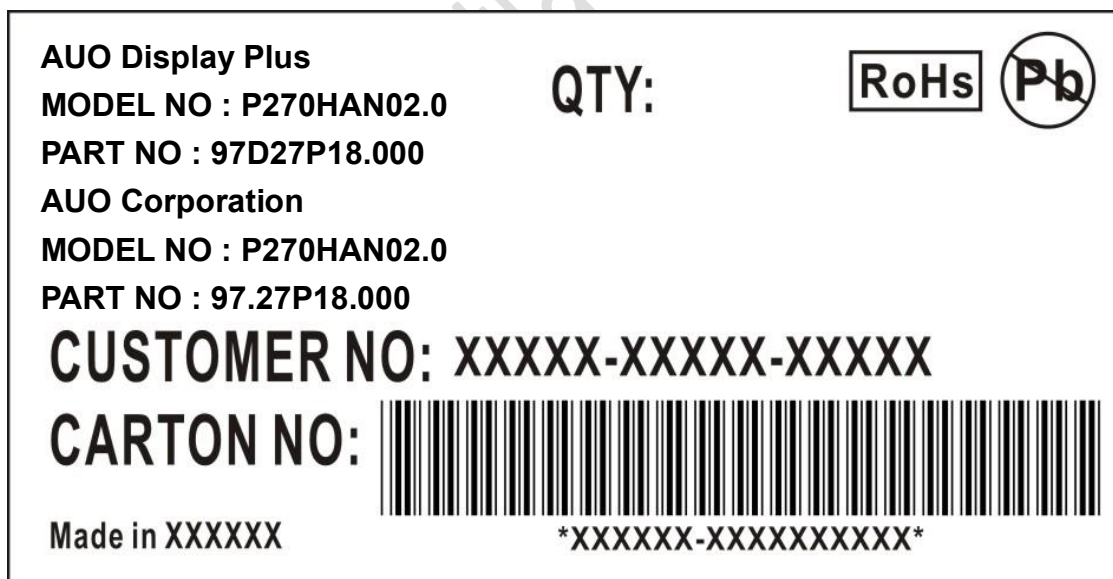
10. Packing

10.1 Definition of Label

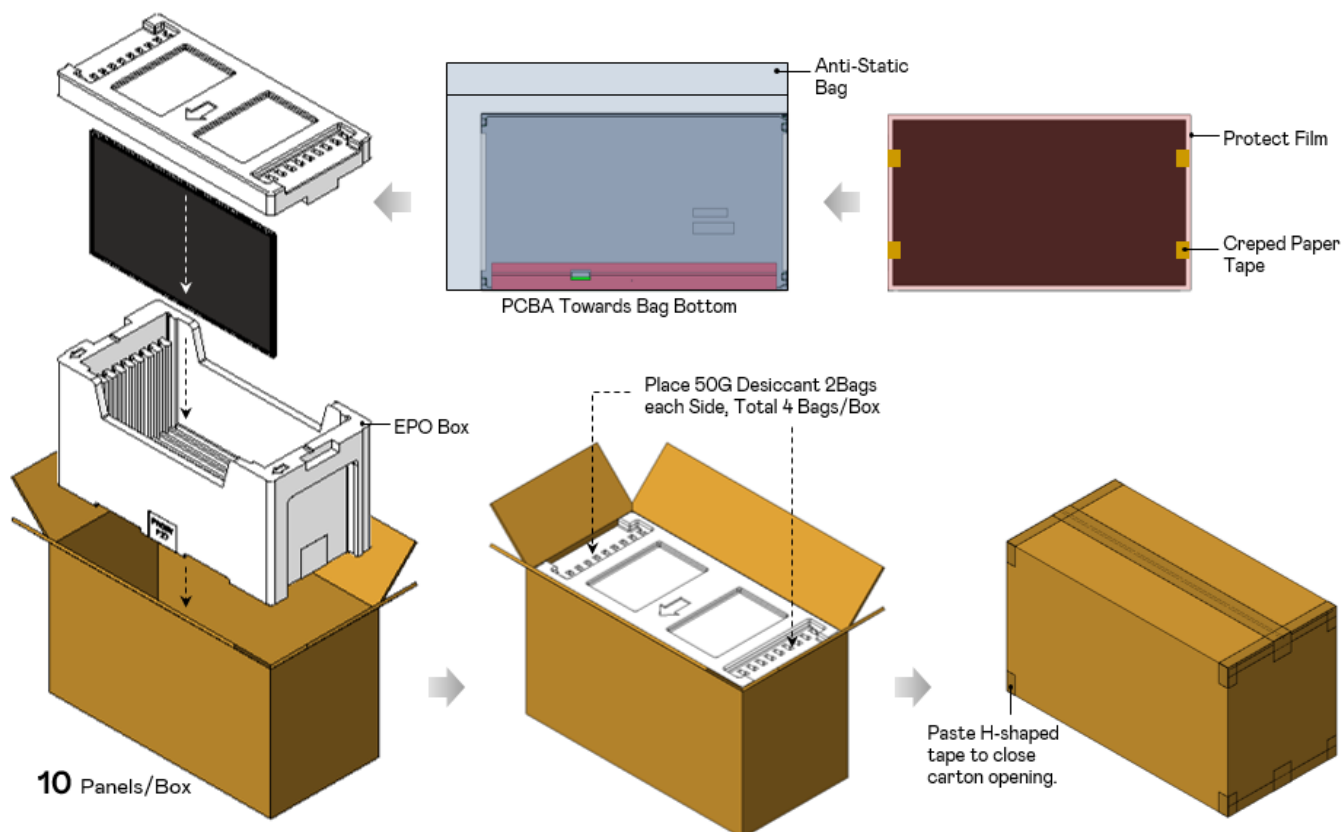
10.1.1 Panel Label



10.1.2 Carton Label



10.2 Packing Methods



10.2.1 Pallet and Shipment Information

Item	Specification			Packing Remark
	Qty.	Dimension	Weight (kg)	
Panel	1	623.7(H)*362.1(W)*12(D)mm	3.233	
Box	1	756(L)*371(W)*467(H)mm	1.57	without panel and cushion
Packing Box	10 Panels/Box	756(L)*371(W)*467(H)mm	35.5	with panel and cushion
Pallet	1	1150(L)*840(W)*132(H)mm	13.6	
Pallet after packing	6 boxes / pallet	1150(L)*840(W)*1066(H)mm	231.7	



11. Precautions

Please pay attention to the followings when you use this TFT LCD module.

11.1 Mounting Precautions

- (1) You must mount a module using holes arranged in four corners or four sides.
- (2) You should consider the mounting structure so that uneven force (ex. twisted stress) is not applied to module. And the case on which a module is mounted should have sufficient strength so that external force is not transmitted directly to the module.
- (3) Please attach the surface transparent protective plate to the surface in order to protect the polarizer. Transparent protective plate should have sufficient strength in order to the resist external force.
- (4) You should adopt radiation structure to satisfy the temperature specification.
- (5) Acetic acid type and chlorine type materials for the cover case are not desirable because the former generates corrosive gas of attacking the polarizer at high temperature and the latter cause circuit broken by electro-chemical reaction.
- (6) Do not touch, push or rub the exposed polarizer with glass, tweezers or anything harder than HB pencil lead. And please do not rub with dust clothes with chemical treatment. Do not touch the surface of polarizer for bare hand or greasy cloth. (Some cosmetics are detrimental to the polarizer.)
- (7) When the surface becomes dusty, please wipe gently with absorbent cotton or other soft materials like chamois soaks with petroleum benzene. Normal-hexane is recommended for cleaning the adhesives used to attach front/ rear polarizer. Do not use acetone, toluene and alcohol because they cause chemical damage to the polarizer.
- (8) Wipe off saliva or water drops as soon as possible. Their long time contact with polarizer causes deformations and color fading.
- (9) Do not open the case because inside circuits do not have sufficient strength.

11.2 Operating Precautions

- (1) The spike noise causes the mis-operation of circuits. It should be lower than following voltage:
 $V = \pm 200\text{mV}$ (Over and under shoot voltage)
- (2) Response time depends on the temperature. (In lower temperature, it becomes longer.)
- (3) Brightness depends on the temperature. (In lower temperature, it may become lower.) And in lower temperature, response time (required time that brightness is stable after turned on) becomes longer.
- (4) Be careful for condensation at sudden temperature change. Condensation makes damage to polarizer or electrical contacted parts. And after fading condensation, smear or spot will occur.
- (5) When fixed patterns are displayed for a long time, remnant image is likely to occur.
- (6) Module has high frequency circuits. Sufficient suppression to the electromagnetic interference shall be done by system manufacturers. Grounding and shielding methods may be important to minimize the interface.

- (7) The conductive material and signal cables are kept away from LED driver inductor to prevent abnormal display, sound noise and temperature rising.

11.3 Operating Condition for Public Information Display

The device listed in the product specification is designed and manufactured for PID (Public Information Display) application. To optimize module's lifetime and function, below operating usages are required.

- (1) Normal operating condition
 - A. Operating temperature: 0~50°C
 - B. Operating humidity: 10~90%
 - C. Display pattern: dynamic pattern (Real display).
Note) Long-term static display would cause image sticking.
- (2) Operation usage to protect against image sticking due to long-term static display.
 - A. Suitable operating time: under 24 hours a day
 - B. Liquid Crystal refresh time is required. Cycling display between 5 minutes' information (static) display and 10 seconds' moving image.
 - C. Periodically change background and character (image) color.
 - D. Avoid combination of background and character with large different luminance.
- (3) Periodically adopt one of the following actions after long time display.
 - A. Running the screen saver (motion picture or black pattern)
 - B. Power off the system for a while
- (4) LCD system is required to place in well-ventilated environment. Adapting active cooling system is highly recommended.
- (5) Product reliability and functions are only guaranteed when the product is used under right operation usages. If product will be used in extreme conditions, such as high temperature/humidity, display stationary patterns, or long operation time etc..., it is strongly recommended to contact ADP for filed application engineering advice. Otherwise, its reliability and function may not be guaranteed. Extreme conditions are commonly found at airports, transit stations, banks, stock market and controlling systems.

11.4 Electrostatic Discharge Control

Since a module is composed of electronic circuits, it is not strong to electrostatic discharge. Make certain that treatment persons are connected to ground through wristband etc. And don't touch interface pin directly.

11.5 Precautions for Strong Light Exposure

- (1) Strong light exposure causes degradation of polarizer and color filter.
- (2) To keep display function well as a digital signage application, especially the component of TFT is very sensitive to sunlight, it is necessary to set up blocking device protecting panel from radiation of ambient environment.

11.6 Storage

When storing modules as spares for a long time, the following precautions are necessary.

- (1) Store them in a dark place. Do not expose the module to sunlight or fluorescent light. Keep the temperature between 5°C and 35°C at normal humidity.
- (2) The polarizer surface should not come in contact with any other object. It is recommended that they be stored in the container in which they were shipped.
- (3) Storage condition is guaranteed under packing conditions.
- (4) The phase transition of Liquid Crystal in the condition of the low or high storage temperature will be recovered when the LCD module returns to the normal condition.

11.7 Handling Precautions for Protection Film

- (1) The protection film is attached to the bezel with a small masking tape. When the protection film is peeled off, static electricity is generated between the film and polarizer. This should be peeled off slowly and carefully by people who are electrically grounded and with well ion-blown equipment or in such a condition, etc.
- (2) When the module with protection film attached is stored for a long time, sometimes there remains a very small amount of glue still on the bezel after the protection film is peeled off.
- (3) You can remove the glue easily. When the glue remains on the bezel or its vestige is recognized, please wipe them off with absorbent cotton waste or other soft material like chamois soaked with normal-hexane.

11.1. Dust Resistance

- (1) ADP module dust tests are conducted with marked areas (e.g., holes and slits around the front bezel and back cover) sealed, to comply with JIS D0207 (see Figure 1).
- (2) To prevent particles from entering the module, please ensure the set has all the highlighted areas (holes and slits) adequately sealed or covered by set mechanism.
- (3) ADP's testing procedure cannot replicate all real world operation scenarios. It is up to the module user to apply the most appropriate dust resistance solution for its particular application.